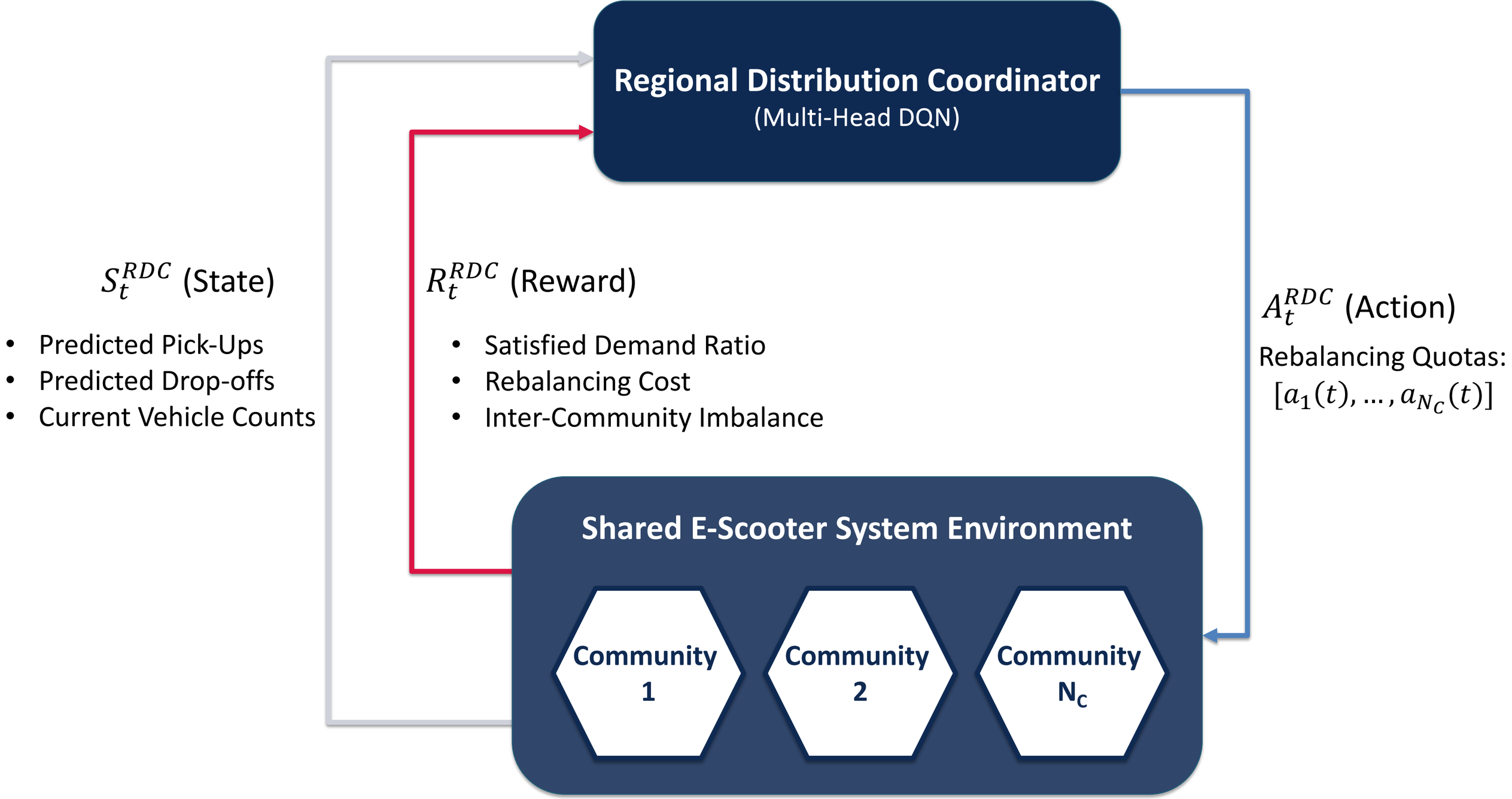
Regional Distribution Coordinator
(Multi-Head DQN)
S_t^{RDC} (State)
Predicted Pick-Ups
Predicted Drop-offs
Current Vehicle Counts
R_t^{RDC} (Reward)
Satisfied Demand Ratio
Rebalancing Cost
Inter-Community Imbalance
A_t^{RDC} (Action)
Rebalancing Quotas:
$[a_1(t), \dots, a_{N_C}(t)]$
Shared E-Scooter System Environment
Community 1
Community 2
Community N_C